

LETTER & DOCUMENT TEMPLATE MANAGEMENT MODULE

Version 2.0 — Final Implementation-Ready Plan

SaaS High School Management System Republic of Liberia

Companion Document to System Architecture v4

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Status: Final — Ready for Implementation

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1. Module Overview

The Letter & Document Template Management Module is a core component of the SaaS High School Management System designed specifically for Liberian secondary schools. It provides a centralized, role-controlled system for creating, approving, delivering, and tracking all official school correspondence.

1.1 Purpose

This module addresses the critical need for standardized, professional school communication in the Liberian educational context. It replaces ad-hoc, inconsistent letter generation with a structured system that ensures:

- Legal compliance through audit trails and tamper-proof records
- Professional branding with school-customized templates
- Role-based access control aligned with the 12 system roles
- Multi-channel delivery suited to Liberia's connectivity challenges
- Offline-first operation consistent with the v4 system architecture

1.2 Scope

The module covers 23 letter types across 7 categories, supporting the full lifecycle from template creation to delivery confirmation and archival. It integrates with Admissions, Discipline, Finance, Attendance, and Communication modules.

1.3 Key Enhancements in v2

Version 2 incorporates 10 critical improvements over v1:

- Bulk operations for mass letter generation and batch printing
- Offline & sync support with IndexedDB draft creation
- Template versioning with rollback and change logs
- Letter recall & void system for error correction
- Enhanced SMS specifications with character limits and URL shortening
- Print queue management dashboard
- Parent acknowledgment and read receipts
- Digital signature verification for high-stakes notices
- Module integration triggers with configurable conditions
- Localization and language support for Liberian context

2. Letter Categories & Types

All 23 letter types are organized into 7 categories. Each letter type has defined severity levels, required approvals, and default delivery channels.

2.1 Complete Letter Type Matrix

Category	Letter Type	Severity	Approval Required	Delivery Channels
Admissions	Acceptance Letter	Low	Registrar	PDF, Portal, Email
Admissions	Rejection Letter	Medium	Principal	PDF, Portal
Admissions	Waitlist Notification	Low	Registrar	PDF, Portal, SMS
Admissions	Transfer Letter	Medium	Principal	PDF
Disciplinary	Warning Letter	Medium	Dean of Students	PDF, Portal
Disciplinary	Suspension Notice	High	Principal	PDF, Portal, SMS
Disciplinary	NTR (Never To Return)	Critical	Principal + Proprietor	PDF only
Disciplinary	Expulsion Notice	Critical	Principal + Proprietor	PDF only
Academic	Report Card Cover Letter	Low	Auto-generated	PDF, Portal
Academic	Promotion Letter	Low	Auto-generated	PDF, Portal
Academic	Retention Notice	High	Principal	PDF, Portal
Academic	Honor Roll Certificate	Low	Auto-generated	PDF, Portal, Email
Financial	Fee Reminder	Medium	Bursar	PDF, SMS, Portal
Financial	Outstanding Balance Notice	High	Bursar + Principal	PDF, SMS
Financial	Payment Receipt	Low	Auto-generated	PDF, Portal, Email
Attendance	Chronic Absenteeism Notice	High	Dean + Principal	PDF, SMS, Portal
Attendance	Truancy Notice	Critical	Principal	PDF, SMS
Communication	PTA Meeting Invitation	Low	Admin Officer	PDF, SMS, Portal, Email
Communication	General Announcement	Low	Admin Officer	Portal, SMS, Email
Communication	Emergency Notice	Critical	Principal	SMS, Portal, Email
Administrative	Enrollment Verification	Low	Registrar	PDF
Administrative	Recommendation Letter	Low	Teacher/Principal	PDF
Administrative	Withdrawal Confirmation	Medium	Registrar + Principal	PDF, Portal

2.2 Severity Levels Defined

Level	Description	Approval Chain	Audit Requirements
Low	Routine communications	Auto or single approver	Standard logging
Medium	Notifications with consequences	Department head	Full audit trail
High	Significant impact on student	Principal required	Full audit + acknowledgment
Critical	Legal implications	Principal + Proprietor review	Full audit + signatures + acknowledgment

3. Template Builder Design

The Template Builder provides school administrators with a visual, drag-and-drop interface for creating and customizing letter templates. Each school can maintain its own branded templates while adhering to system-enforced compliance requirements.

3.1 Template Components

- **School Logo:** Upload and position school logo (PNG/JPG, max 2MB, recommended 300x300px)
- **Header Block:** School name, address, motto, contact information
- **Letter Body:** Rich text editor with placeholder insertion toolbar
- **Signature Block:** Signatory name, title, and signature image placement
- **Footer:** School registration number, accreditation info, confidentiality notice
- **Watermark:** Optional school seal or “OFFICIAL” watermark overlay

3.1.1 Digital Signature System

High-stakes letters (NTR, Expulsion, Suspension) require verified digital signatures:

- Each authorized signatory uploads a signature image (PNG with transparent background)
- Signature placement is defined per template with X/Y coordinates and size
- Signatory name and title auto-populate below the signature image
- Multi-signature support: NTR and Expulsion letters require both Principal and Proprietor signatures
- Signature audit trail: system logs which signature was applied, by whom, timestamp, and IP address
- Signature images are encrypted at rest and only rendered during PDF generation

3.2 Template Customization Options

Element	Customizable Properties	Constraints
Logo	Position, size, opacity	Max 2MB, PNG/JPG only
Header	Font, size, color, alignment	School name required
Body Text	Font family, size, line spacing	Min 10pt, max 14pt
Margins	Top, bottom, left, right	Min 0.5 inch all sides
Signature	Position, size, name format	Must match authorized signatory
Footer	Content, font size	Registration number required
Watermark	Text or image, opacity	Max 30% opacity

3.3 Bulk Operations

The system supports mass letter generation for entire grades, classes, or filtered student groups. This is essential for operations like sending fee reminders to all students with outstanding balances.

3.3.1 Batch Generation Workflow

- Admin selects a letter template and defines the recipient filter (grade, class, status, balance threshold, etc.)
- System displays a preview count: “200 students match this filter”
- Admin confirms generation; system queues all letters as a background job
- Progress bar shows real-time status: “142/200 generated”
- Failed generations are logged with error reasons (e.g., missing placeholder data)
- Completed batch produces a downloadable ZIP of all PDFs or routes to print queue

3.3.2 Batch Generation Limits

Parameter	Limit	Rationale
Max batch size	500 letters per job	Server memory constraints
Concurrent jobs	2 per school	Fair resource allocation
Retry attempts	3 per failed letter	Network/data resilience
Job expiry	24 hours	Auto-cleanup of stale jobs

3.3.3 CSV Export

Every bulk operation generates a CSV summary report containing:

- Student name, ID, grade, class
- Letter generation status (Success/Failed/Skipped)
- Delivery channel used
- Error details for failed generations
- Timestamp of generation

3.4 Template Versioning & Governance

Every modification to a letter template creates a new version, ensuring full traceability and the ability to roll back to any previous version.

3.4.1 Version Control Rules

- Each template edit creates an incremental version (v1, v2, v3...)
- Only one version can be “Active” at any time — this is the version used for letter generation
- Edits are saved as “Draft” versions until explicitly promoted to Active
- Rollback: any previous version can be promoted to Active with one click
- Version comparison: side-by-side diff view showing changes between any two versions

- Change log: every version records who edited, what changed, and when

3.4.2 Template Version Schema

Field	Type	Description
version_id	UUID	Unique version identifier
template_id	UUID (FK)	Parent template reference
version_number	INTEGER	Sequential version number
content_json	JSONB	Full template content snapshot
status	ENUM	draft active archived
created_by	UUID (FK)	User who created this version
created_at	TIMESTAMP	Version creation timestamp
change_summary	TEXT	Human-readable change description

3.5 Localization & Language Support

Templates support localization for the Liberian context:

- **Primary Language:** English (standard)
- **Alternate Variant:** Simple Liberian English for parent-facing documents
- **Date Format:** DD/MM/YYYY (Liberian standard), with {{date_formatted}} placeholder
- **Currency:** Dual display — USD and LRD with {{amount_usd}} and {{amount_lrd}} placeholders
- **Exchange Rate:** Configurable exchange rate placeholder {{exchange_rate}} updated by Bursar
- Template variants: each template can have multiple language versions
- Auto-selection: system selects variant based on parent's preferred language setting

4. Dynamic Placeholders System

Placeholders are variables embedded in templates that auto-fill with real data during letter generation. The system validates that all required placeholders have data before generating a letter.

4.1 Complete Placeholder Reference

Category	Placeholder	Description	Used In
Student	{{student_name}}	Full name of the student	All
Student	{{student_id}}	Unique student ID number	All
Student	{{student_grade}}	Current grade level	Academic, Attendance
Student	{{student_class}}	Class/section assignment	Academic, Attendance
Student	{{student_dob}}	Date of birth (DD/MM/YYYY)	Administrative
Student	{{student_gender}}	Gender (Male/Female)	Administrative
Parent	{{parent_name}}	Parent/guardian full name	All
Parent	{{parent_phone}}	Primary phone number	Communication
Parent	{{parent_email}}	Email address (if available)	Communication
Parent	{{parent_address}}	Mailing address	All physical letters
School	{{school_name}}	Official school name	All
School	{{school_address}}	School physical address	All
School	{{school_phone}}	School contact number	All
School	{{school_motto}}	School motto	Headers
School	{{school_reg_number}}	MOE registration number	Official letters
Date	{{date}}	Current date (DD/MM/YYYY)	All
Date	{{date_formatted}}	Formatted date (e.g., 8 April 2026)	All
Date	{{academic_year}}	Current academic year	Academic
Date	{{semester}}	Current semester/term	Academic
Finance	{{fee_amount}}	Total fee amount due	Financial
Finance	{{amount_paid}}	Amount already paid	Financial
Finance	{{outstanding_balance}}	Remaining balance	Financial
Finance	{{payment_deadline}}	Payment due date	Financial
Finance	{{amount_usd}}	Amount in US Dollars	Financial
Finance	{{amount_lrd}}	Amount in Liberian Dollars	Financial
Finance	{{exchange_rate}}	Current USD/LRD exchange rate	Financial
Discipline	{{incident_date}}	Date of incident	Disciplinary
Discipline	{{incident_description}}	Description of the offense	Disciplinary

Discipline	{{disciplinary_action}}	Action taken	Disciplinary
Discipline	{{suspension_start}}	Suspension start date	Disciplinary
Discipline	{{suspension_end}}	Suspension end date	Disciplinary
Discipline	{{return_conditions}}	Conditions for return	Disciplinary
Attendance	{{days_absent}}	Total days absent	Attendance
Attendance	{{absence_percentage}}	Absence rate percentage	Attendance
Attendance	{{last_attendance_date}}	Last date student attended	Attendance
Signatory	{{signatory_name}}	Name of letter signer	All
Signatory	{{signatory_title}}	Title of letter signer	All
Signatory	{{signatory_signature}}	Digital signature image	High/Critical
System	{{letter_reference}}	Unique letter reference number	All
System	{{letter_date}}	Date letter was generated	All
System	{{verification_code}}	QR/code for letter verification	Critical

4.2 Placeholder Validation

Before generating any letter, the system performs validation:

- Required placeholders: the system checks that all placeholders in the template have corresponding data
- Missing data handling: if a required placeholder has no data, generation is blocked with an error message
- Optional placeholders: marked with {{?placeholder}} syntax; rendered as blank if no data
- Preview mode: admin can see a preview with sample data before sending
- Data type validation: dates must be valid dates, amounts must be numeric, etc.

5. Delivery Channels

The system provides four delivery channels designed for Liberia's mixed-connectivity environment. Schools configure default channels per letter type, with manual override available.

5.1 Channel Overview

Channel	Primary Use	Connectivity Required	Cost	Reliability
PDF Download/Print	All letter types	Generation only	Paper + ink	Highest
In-App (Parent Portal)	Non-critical notifications	Internet access	Free	Medium
SMS Link	Urgent and financial notices	Basic cellular	Per-message fee	High
Email	Where available	Internet access	Free	Low (limited adoption)

5.2 PDF Download & Print

PDF is the primary delivery channel, reflecting Liberia's reliance on physical document delivery.

- **Generation:** Server-side PDF rendering using Puppeteer or wkhtmltopdf
- **Format:** A4 size, portrait orientation, with school branding
- **Storage:** Generated PDFs stored in object storage with 7-year retention
- **Download:** Individual or bulk download (ZIP for batch operations)

5.2.1 Print Queue Management

The Print Queue Dashboard provides centralized management of all letters awaiting physical printing and distribution:

- **Queue Status Board:** Visual dashboard showing Pending, Printing, Printed, Distributed, Failed
- **Batch Printing:** Group letters by grade, class, or letter type for efficient printing
- **Page Count Estimator:** Shows total pages before printing to manage paper/ink resources
- **Priority Queue:** Critical-severity letters automatically promoted to top of queue

5.2.2 Physical Distribution Tracking

Status	Description	Set By
Queued	Letter generated, awaiting print	System (auto)
Printing	Currently being printed	Admin (manual)
Printed	Printed, awaiting distribution	Admin (manual)
Handed to Student	Given directly to student	Admin/Teacher

Mailed	Sent via postal service	Admin
Picked Up	Parent collected from office	Admin
Failed	Print error occurred	System (auto)

- Reprint: any letter can be reprinted with audit log entry recording reason
- Distribution report: summary of delivery methods used per batch

5.3 In-App Notification (Parent Portal)

- Letter appears in parent's portal inbox with unread indicator
- Push notification triggered (if parent has mobile app installed)
- Full letter viewable in-app with option to download PDF
- Read receipt: system records when parent opens the letter

5.4 SMS Link

SMS delivery is critical for reaching parents without smartphones or internet access.

5.4.1 SMS Message Format

SMS messages must fit within 160 characters to avoid multi-part charges:

Letter Type	SMS Template (max 160 chars)	Character Count
Fee Reminder	[SchoolName]: Fee reminder for [Student]. Balance: \$ [amt]. View: [short_url]	~85
Suspension	[SchoolName]: [Student] suspended [start]-[end]. Details: [short_url]	~75
Emergency	[SchoolName] URGENT: [brief_msg]. Check portal or call [phone]	~70
PTA Meeting	[SchoolName]: PTA meeting [date] at [time]. Details: [short_url]	~65
Attendance	[SchoolName]: [Student] absent [X] days. Please contact school. [short_url]	~80

5.4.2 URL Shortening

- System generates short URLs (e.g., sch.ly/abc123) linking to full letter in parent portal
- Short URLs are unique per letter instance and expire after 90 days
- Fallback: if URL shortener service is unavailable, send summary-only SMS without link
- Click tracking: system logs when a short URL is accessed

5.4.3 SMS Gateway Integration

- Primary gateway: configurable per school (e.g., Twilio, Africa's Talking, local Liberian providers)
- Delivery status callbacks: Sent, Delivered, Failed, Undeliverable

- Retry logic: failed SMS retried up to 3 times at 15-minute intervals
- Cost tracking: per-school SMS usage dashboard with monthly spend

5.5 Email

- HTML-formatted email with letter content and PDF attachment
- Fallback plain-text version for older email clients
- Delivery tracking via read receipts and bounce handling
- Email is optional and supplementary given low email adoption in Liberia

5.6 Offline & Sync Support

Consistent with the v4 system's offline-first architecture, the letter module supports full offline operation:

5.6.1 Offline Draft Creation

- Teachers and admins can draft letters while offline using IndexedDB storage
- Template data and placeholder values are cached locally during last sync
- Drafts are marked with "Offline Draft" status and timestamped
- Rich text editing available offline using cached template components

5.6.2 Sync Queue

- When connectivity returns, offline drafts enter a sync queue
- Sync order: oldest drafts first (FIFO)
- Conflict resolution: if the same letter was edited both offline and online, server version wins; offline version is preserved as "Draft Copy"
- Sync status indicators: Pending Sync, Syncing, Synced, Conflict
- Failed syncs retry automatically up to 5 times

5.6.3 Offline PDF Viewing

- Previously generated PDFs are cached in the browser for offline viewing
- Cache limit: 50 most recent PDFs per user
- Cache is refreshed on each successful sync

6. Approval Workflow

Letters follow role-based approval chains mapped to the 12 system roles. The approval requirements vary by letter severity.

6.1 Approval Chain by Severity

Severity	Draft By	Approval Required	Time Limit	Escalation
Low	Any authorized role	None (auto-approved)	N/A	N/A
Medium	Teacher, Admin Officer	Department Head	24 hours	Auto-approve after 48h
High	Dean, Department Head	Principal	12 hours	Notify Proprietor after 24h
Critical	Principal	Principal + Proprietor Review	Immediate	Cannot bypass

6.2 Proprietor Review (Read-Only Ethics)

In alignment with the v4 system's Proprietor role restrictions, the Proprietor has a special review workflow for critical letters:

- Proprietor receives a notification when a Critical letter is drafted
- Proprietor can VIEW the letter and its context but CANNOT edit content
- Proprietor can add a "Review Note" with comments or concerns
- Proprietor clicks "Acknowledge" to confirm they have reviewed the letter
- If Proprietor has concerns, they flag it and the letter returns to Principal for revision
- This preserves operational authority with the Principal while ensuring oversight

6.3 Approval Workflow States

State	Description	Actions Available
Draft	Letter created, not submitted	Edit, Delete, Submit for Approval
Pending Approval	Submitted, awaiting approver	Approve, Reject, Request Changes
Changes Requested	Returned with feedback	Edit and Resubmit
Approved	Ready for delivery	Send, Schedule, Add to Print Queue
Sent	Delivered via selected channel(s)	View, Reprint, Track Delivery
Recalled	Withdrawn after sending	View Recall Reason
Voided	Marked as invalid	View (with VOID watermark)

6.4 Letter Recall & Void

The system provides mechanisms to handle letters sent in error:

6.4.1 Recall System

- **Time Window:** Letters can be recalled within a configurable window (default: 2 hours after sending)
- **Recall Actions:**
 - If delivered via portal: letter is removed from parent’s inbox with “Recalled” notice
 - If delivered via email: a follow-up “Retraction Notice” email is auto-sent
 - If delivered via SMS: a follow-up SMS is sent: “Please disregard previous message from [School]”
 - If printed/distributed: physical recall is logged manually by admin
- **Recall Reason:** Admin must provide a reason (Sent in Error, Wrong Recipient, Incorrect Information, Other)

6.4.2 Void System

- **Purpose:** For letters that cannot be recalled but need to be invalidated in the system
- Voided letters are NEVER deleted — they remain in the audit trail
- Voided letters display a “VOID” watermark if reprinted or viewed
- Void requires Principal-level authority
- Void reason and approver are permanently logged

6.4.3 Recall/Void Schema

Field	Type	Description
recall_id	UUID	Unique recall record identifier
letter_instance_id	UUID (FK)	Reference to the recalled letter
action_type	ENUM	recall void
reason	ENUM	sent_in_error wrong_recipient incorrect_info other
reason_detail	TEXT	Free-text explanation
initiated_by	UUID (FK)	User who initiated the action
approved_by	UUID (FK)	User who approved (for void)
created_at	TIMESTAMP	When the action was taken
channels_notified	JSONB	Which delivery channels were notified

7. Integration Points

The Letter Module integrates with all major system modules. Each integration includes configurable auto-triggers with conditions and suppression rules.

7.1 Module Integration Triggers

Source Module	Trigger Event	Letter Generated	Condition	Delay
Admissions	Application approved	Acceptance Letter	Always	Immediate
Admissions	Application rejected	Rejection Letter	Always	Immediate
Admissions	Waitlist placement	Waitlist Notification	Always	Immediate
Admissions	Transfer processed	Transfer Letter	Always	Immediate
Discipline	Warning recorded	Warning Letter	Always	1 hour
Discipline	Suspension entered	Suspension Notice	Always	Immediate
Discipline	NTR decision	NTR Notice	Principal approved	After Proprietor review
Discipline	Expulsion decision	Expulsion Notice	Principal approved	After Proprietor review
Finance	Balance exceeds threshold	Fee Reminder	Balance > \$50 USD	3 days after threshold
Finance	Payment received	Payment Receipt	Always	Immediate
Finance	30 days overdue	Outstanding Balance	Balance > \$0	Immediate
Attendance	10+ days absent	Absenteeism Notice	Consecutive or cumulative	Next school day
Attendance	Truancy threshold	Truancy Notice	Per MOE guidelines	Immediate
Communication	PTA scheduled	PTA Invitation	Always	7 days before event
System	Report cards generated	Report Card Cover	End of term	With report cards
Academic	Promotion decided	Promotion Letter	End of year	With results
Academic	Retention decided	Retention Notice	End of year	With results

7.2 Trigger Configuration

Administrators can configure trigger behavior per school:

- **Enable/Disable:** Any auto-trigger can be turned off; letters must then be created manually

- **Delay Configuration:** Adjustable delay between trigger event and letter generation (0 mins to 30 days)
- **Condition Thresholds:** Numeric thresholds (e.g., fee balance, absent days) are configurable per school
- **Suppression Rules:** Prevent duplicate notifications:
 - Same letter type to same recipient within X days (configurable, default: 7 days)
 - Parent already contacted via phone/meeting this week
 - Student has pending unacknowledged letter of same type
- **Manual Override:** Admin can force-trigger or suppress any individual auto-trigger

7.3 Data Flow Diagram

Integration data flows through a centralized Event Bus:

- Source module emits an event (e.g., "suspension.created")
- Event Bus checks trigger configuration for the school
- If trigger is active and conditions met, Letter Service is invoked
- Letter Service pulls template, fills placeholders from event data
- Approval workflow is initiated based on letter severity
- Upon approval, delivery is executed via configured channels

8. Audit Trail & Compliance

Every letter action is logged in an immutable audit trail for legal protection and regulatory compliance. This is especially critical for disciplinary letters that may face legal challenge.

8.1 Audit Log Fields

Field	Type	Description
audit_id	UUID	Unique audit entry identifier
letter_instance_id	UUID (FK)	Reference to the letter
action	ENUM	created submitted approved rejected sent delivered read recalled voided reprinted acknowledged
performed_by	UUID (FK)	User who performed the action
performed_at	TIMESTAMP	Exact timestamp of the action
ip_address	VARCHAR	IP address of the user
user_agent	TEXT	Browser/device information
metadata	JSONB	Additional context (delivery channel, error details, etc.)
checksum	VARCHAR	SHA-256 hash for tamper detection

8.2 Compliance Requirements

- **Retention Period:** All letter records and audit logs retained for minimum 7 years
- **Tamper Protection:** Audit entries are append-only; no update or delete operations permitted
- **Checksum Verification:** Each audit entry includes a SHA-256 hash of the previous entry, creating a chain
- **Export:** Full audit trail exportable as CSV or PDF for legal proceedings
- **Access:** Only Principal and Proprietor can view full audit trails; other roles see their own actions only

8.3 Parent Acknowledgment & Read Receipts

For High and Critical severity letters, the system tracks whether parents have received and acknowledged the communication:

8.3.1 Acknowledgment Methods

Method	How It Works	Applicable When
Digital (Portal)	Parent clicks “I have read this letter” button	Parent has portal access
Digital (SMS)	Parent replies “YES” to acknowledgment SMS	SMS delivery channel used

Physical Sign-off	Admin marks "Parent signed physical copy"	Letter delivered by hand
Phone Confirmation	Admin records "Confirmed via phone call on [date]"	Follow-up call made

8.3.2 Escalation Rules

- Day 1: Letter sent via configured channels
- Day 3 (configurable): If no acknowledgment, auto-send follow-up via SMS
- Day 7 (configurable): If still no acknowledgment, notify Dean/Principal
- Day 14 (configurable): Final notice sent; case flagged for in-person follow-up
- All escalation steps are logged in the audit trail

8.3.3 Acknowledgment Dashboard

A dedicated dashboard shows acknowledgment rates:

- Acknowledgment rate by letter type (e.g., 85% of fee reminders acknowledged)
- Outstanding acknowledgments by grade/class
- Average time to acknowledgment
- Parents who have never acknowledged any letter (for follow-up)

8.3.4 Acknowledgment Schema

Field	Type	Description
acknowledgment_id	UUID	Unique identifier
letter_instance_id	UUID (FK)	Reference to the letter
acknowledged_by	UUID (FK)	Parent user ID
method	ENUM	digital_portal digital_sms physical_signoff phone_confirmation
acknowledged_at	TIMESTAMP	When acknowledgment was recorded
recorded_by	UUID (FK)	Admin who recorded (for physical/phone)
notes	TEXT	Additional notes from admin
signature_data	TEXT	Base64 signature image (for physical)

9. Starter Templates

The system ships with pre-built starter templates for all 23 letter types. Schools can use these immediately or customize them with their own branding.

9.1 Starter Template List

Category	Template Name	Placeholders Used	Default Channel
Admissions	Acceptance Letter	student_name, parent_name, grade, academic_year, date	PDF + Portal
Admissions	Rejection Letter	student_name, parent_name, date	PDF
Admissions	Waitlist Notification	student_name, parent_name, date, waitlist_position	PDF + SMS
Admissions	Transfer Letter	student_name, parent_name, school_name, date	PDF
Disciplinary	Warning Letter	student_name, parent_name, incident_date, incident_description	PDF + Portal
Disciplinary	Suspension Notice	student_name, parent_name, suspension_start, suspension_end, incident_description	PDF + SMS
Disciplinary	NTR Notice	student_name, parent_name, incident_description, signatory_name, signatory_title	PDF only
Disciplinary	Expulsion Notice	student_name, parent_name, incident_description, disciplinary_action	PDF only
Academic	Report Card Cover	student_name, grade, class, semester, academic_year	PDF
Academic	Promotion Letter	student_name, parent_name, grade, academic_year	PDF + Portal
Academic	Retention Notice	student_name, parent_name, grade, academic_year	PDF + Portal
Academic	Honor Roll Certificate	student_name, grade, semester, academic_year	PDF + Portal
Financial	Fee Reminder	student_name, parent_name, fee_amount, outstanding_balance, payment_deadline	PDF + SMS
Financial	Outstanding Balance	student_name, parent_name, outstanding_balance, amount_usd, amount_lrd	PDF + SMS
Financial	Payment Receipt	student_name, parent_name, amount_paid, date, letter_reference	PDF + Portal
Attendance	Absenteeism Notice	student_name, parent_name, days_absent, absence_percentage	PDF + SMS

Attendance	Truancy Notice	student_name, parent_name, days_absent, last_attendance_date	PDF + SMS
Communication	PTA Meeting Invitation	parent_name, date, school_name	SMS + Portal
Communication	General Announcement	school_name, date	Portal + SMS
Communication	Emergency Notice	school_name, date	SMS + Portal
Administrative	Enrollment Verification	student_name, student_id, grade, academic_year	PDF
Administrative	Recommendation Letter	student_name, grade, signatory_name, signatory_title	PDF
Administrative	Withdrawal Confirmation	student_name, parent_name, date, grade	PDF + Portal

9.2 Sample Template Structure

Each starter template follows this standard structure:

- [School Logo] — positioned top-left or top-center
- [School Name & Address Header] — full width, centered
- [Date & Reference Number] — right-aligned
- [Recipient Address Block] — parent name and address
- [Subject Line] — bold, underlined
- [Salutation] — “Dear {{parent_name}},”
- [Body Paragraphs] — template-specific content with placeholders
- [Action Required Block] — highlighted box for required parent actions (if any)
- [Signature Block] — signatory name, title, and signature image
- [Footer] — school registration, confidentiality notice

10. Role Permissions for Letters

Permissions are mapped across all 12 system roles. Each role has specific draft, approve, and send permissions.

10.1 Permission Matrix

Role	Draft	Approve	Send	View	Manage Templates
Proprietor	✗	Review only (Critical)	✗	✓	✓
Principal	✓	✓ (all levels)	✓	✓	✓
Vice Principal	✓	✓ (up to High)	✓	✓	✓
Dean of Students	✓	✓ (up to Medium)	✓	✓	✗
Registrar	✓	✗	✓ (Admissions)	✓	✗
Bursar	✓	✗	✓ (Financial)	✓	✗
Department Head	✓	✓ (Low)	✓ (Academic)	✓	✗
Teacher	✓ (limited)	✗	✗	✓ (own)	✗
Admin Officer	✓	✗	✓ (Communication)	✓	✗
IT Administrator	✗	✗	✗	✓	✓
Counselor	✓ (limited)	✗	✗	✓ (own)	✗
Parent	✗	✗	✗	✓ (received)	✗

10.2 Special Permission Rules

- **Proprietor:** Cannot draft or send letters directly. Can only review Critical letters and view audit trails. This enforces the read-only ethics principle from v4.
- **Teacher:** Can draft recommendation letters and academic notices for their own students only. Cannot send without Department Head approval.
- **Parent:** Can only view letters addressed to them. Can acknowledge receipt but cannot draft, approve, or send.
- **IT Administrator:** Can manage templates and system configuration but cannot draft or send letters. Has full access to audit logs for system maintenance.

11. Technical Schema

Complete database schema for the Letter Module, including all v2 enhancements.

11.1 Core Tables

11.1.1 letter_templates

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Template unique ID
school_id	UUID	FK → schools(id), NOT NULL	Owning school
name	VARCHAR(255)	NOT NULL	Template display name
category	ENUM	NOT NULL	admissions disciplinary academic financial attendance communication administrative
letter_type	VARCHAR(100)	NOT NULL	Specific letter type
severity	ENUM	NOT NULL	low medium high critical
content_json	JSONB	NOT NULL	Template structure and content
placeholders	JSONB	NOT NULL	Required and optional placeholders
default_channels	JSONB	NOT NULL	Default delivery channels
approval_chain	JSONB	NOT NULL	Required approval roles
active_version_id	UUID	FK → letter_template_versions(id)	Currently active version
is_starter	BOOLEAN	DEFAULT false	System-provided template
created_by	UUID	FK → users(id)	Creator
created_at	TIMESTAMP	DEFAULT NOW()	Creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Last update

11.1.2 letter_instances

Column	Type	Constraints	Description
id	UUID	PK	Letter instance unique ID
template_id	UUID	FK → letter_templates(id)	Source template
template_version_id	UUID	FK → letter_template_versions(id)	Template version used
school_id	UUID	FK → schools(id)	School
student_id	UUID	FK → students(id)	Target student

recipient_data	JSONB	NOT NULL	Filled placeholder values
generated_pdf_url	TEXT		URL to generated PDF
status	ENUM	NOT NULL	draft pending_approval changes_requested approved sent recalled voided
delivery_channels	JSONB		Channels used for delivery
delivery_status	JSONB		Per-channel delivery status
batch_id	UUID	NULLABLE	Batch reference for bulk ops
reference_number	VARCHAR(50)	UNIQUE	Human-readable reference
created_by	UUID	FK → users(id)	Drafter
approved_by	UUID	FK → users(id)	Approver
sent_at	TIMESTAMP		When sent
created_at	TIMESTAMP	DEFAULT NOW()	Creation timestamp

11.2 v2 Enhancement Tables

11.2.1 letter_template_versions

Column	Type	Constraints	Description
id	UUID	PK	Version unique ID
template_id	UUID	FK → letter_templates(id), NOT NULL	Parent template
version_number	INTEGER	NOT NULL	Sequential version (1, 2, 3...)
content_json	JSONB	NOT NULL	Full template content snapshot
status	ENUM	NOT NULL	draft active archived
change_summary	TEXT		Description of changes
created_by	UUID	FK → users(id)	Version author
created_at	TIMESTAMP	DEFAULT NOW()	Version timestamp

11.2.2 letter_recalls

Column	Type	Constraints	Description
id	UUID	PK	Recall record ID
letter_instance_id	UUID	FK → letter_instances(id), NOT NULL	Recalled letter
action_type	ENUM	NOT NULL	recall void

reason	ENUM	NOT NULL	sent_in_error wrong_recipient incorrect_info other
reason_detail	TEXT		Detailed explanation
initiated_by	UUID	FK → users(id)	Who initiated
approved_by	UUID	FK → users(id)	Who approved (for void)
channels_notified	JSONB		Channels that received retraction
created_at	TIMESTAMP	DEFAULT NOW()	Action timestamp

11.2.3 letter_acknowledgments

Column	Type	Constraints	Description
id	UUID	PK	Acknowledgment ID
letter_instance_id	UUID	FK → letter_instances(id), NOT NULL	Acknowledged letter
acknowledged_by	UUID	FK → users(id)	Parent user
method	ENUM	NOT NULL	digital_portal digital_sms physical_signoff phone_confirmation
acknowledged_at	TIMESTAMP	DEFAULT NOW()	When acknowledged
recorded_by	UUID	FK → users(id)	Admin (for physical)
notes	TEXT		Additional notes
signature_data	TEXT		Base64 signature image

11.2.4 print_queue

Column	Type	Constraints	Description
id	UUID	PK	Queue entry ID
letter_instance_id	UUID	FK → letter_instances(id), NOT NULL	Letter to print
school_id	UUID	FK → schools(id)	School
status	ENUM	NOT NULL	queued printing printed distributed failed
priority	INTEGER	DEFAULT 0	Higher = more urgent
distribution_method	ENUM		handed_to_student mailed picked_up
distributed_at	TIMESTAMP		When physically distributed
distributed_by	UUID	FK → users(id)	Who distributed
page_count	INTEGER		Number of pages

reprint_count	INTEGER	DEFAULT 0	Times reprinted
created_at	TIMESTAMP	DEFAULT NOW()	Queue entry timestamp

11.2.5 letter_audit_log

Column	Type	Constraints	Description
id	UUID	PK	Audit entry ID
letter_instance_id	UUID	FK → letter_instances(id)	Related letter
action	ENUM	NOT NULL	created submitted approved rejected sent delivered read recalled voided reprinted acknowledged
performed_by	UUID	FK → users(id)	Actor
performed_at	TIMESTAMP	DEFAULT NOW()	Timestamp
ip_address	VARCHAR(45)		IP address
user_agent	TEXT		Browser/device info
metadata	JSONB		Additional context
checksum	VARCHAR(64)		SHA-256 tamper-detection hash

12. API Endpoints

RESTful API endpoints for all Letter Module operations.

12.1 Template Management

Method	Endpoint	Description	Auth Required
GET	/api/v1/templates	List all templates for school	Authenticated
GET	/api/v1/templates/:id	Get template details	Authenticated
POST	/api/v1/templates	Create new template	Admin+
PUT	/api/v1/templates/:id	Update template (creates new version)	Admin+
GET	/api/v1/templates/:id/versions	List all versions	Admin+
POST	/api/v1/templates/:id/versions/:vid/activate	Set version as active	Admin+
POST	/api/v1/templates/:id/versions/:vid/rollback	Rollback to version	Admin+
DELETE	/api/v1/templates/:id	Soft-delete template	IT Admin+

12.2 Letter Operations

Method	Endpoint	Description	Auth Required
POST	/api/v1/letters	Create/draft a letter	Role-based
POST	/api/v1/letters/bulk	Bulk generate letters	Admin+
GET	/api/v1/letters/:id	Get letter details	Role-based
POST	/api/v1/letters/:id/submit	Submit for approval	Drafter
POST	/api/v1/letters/:id/approve	Approve letter	Approver role
POST	/api/v1/letters/:id/reject	Reject with feedback	Approver role
POST	/api/v1/letters/:id/send	Send via channels	Authorized sender
POST	/api/v1/letters/:id/recall	Recall letter	Principal+
POST	/api/v1/letters/:id/void	Void letter	Principal+
GET	/api/v1/letters/:id/pdf	Download generated PDF	Authorized viewer

12.3 Acknowledgment & Tracking

Method	Endpoint	Description	Auth Required
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POST	/api/v1/letters/:id/acknowledge	Parent acknowledges letter	Parent
GET	/api/v1/acknowledgments	Get acknowledgment dashboard data	Admin+
POST	/api/v1/letters/:id/acknowledge/physical	Record physical sign-off	Admin
GET	/api/v1/letters/:id/delivery-status	Get delivery status per channel	Authorized viewer

12.4 Print Queue

Method	Endpoint	Description	Auth Required
GET	/api/v1/print-queue	List print queue items	Admin+
POST	/api/v1/print-queue/:id/status	Update print status	Admin
POST	/api/v1/print-queue/:id/distribute	Mark as distributed	Admin/Teacher
POST	/api/v1/print-queue/:id/reprint	Reprint a letter	Admin
GET	/api/v1/print-queue/stats	Queue statistics	Admin+

12.5 Audit

Method	Endpoint	Description	Auth Required
GET	/api/v1/audit/letters/:id	Get audit trail for a letter	Principal/Proprietor
GET	/api/v1/audit/export	Export audit logs (CSV/PDF)	Principal/Proprietor
GET	/api/v1/audit/search	Search audit logs with filters	Principal/Proprietor

13. Implementation Roadmap

Recommended phased implementation approach:

13.1 Phase 1: Foundation (Weeks 1–4)

- Database schema creation (all tables)
- Template CRUD operations
- Basic letter generation with placeholder filling
- PDF generation engine
- Starter templates loaded for all 23 types

13.2 Phase 2: Workflow & Delivery (Weeks 5–8)

- Approval workflow engine
- In-app notification delivery
- Email delivery with attachments
- SMS integration with gateway
- Print queue management

13.3 Phase 3: Advanced Features (Weeks 9–12)

- Bulk operations and batch generation
- Template versioning and rollback
- Parent acknowledgment system
- Digital signature integration
- Recall and void functionality

13.4 Phase 4: Integration & Polish (Weeks 13–16)

- Module integration triggers (auto-generation)
- Offline draft creation and sync
- Audit trail with checksum verification
- Localization (date format, dual currency)
- Acknowledgment dashboard and analytics
- Performance optimization and load testing

14. Summary

This document provides a complete, implementation-ready plan for the Letter & Document Template Management Module. It covers:

- 23 letter types across 7 categories with defined severity levels
- Visual template builder with school branding and digital signatures
- 40+ dynamic placeholders with validation rules
- 4 delivery channels optimized for Liberia's connectivity challenges
- Role-based approval workflows aligned with 12 system roles
- Bulk operations for mass letter generation and batch printing
- Offline-first operation with IndexedDB drafts and sync queues
- Template versioning with rollback capability
- Letter recall and void for error correction
- Parent acknowledgment tracking with escalation rules
- Immutable audit trail with SHA-256 tamper detection
- Complete database schema with 6 tables
- RESTful API with 25+ endpoints
- 16-week phased implementation roadmap

— *End of Document* —